

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME

COVERED CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Code of Conduct:

I KONKALA REDDI KISHORE REDDY (192312523)
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Research Article Review – Decision Tree Based Machine Learning for COVID-19 Mortality Classification

Selected Topic

Decision Trees / Inductive Learning

Article Citation Details

Authors: Samad Moslehi, Niloofar Rabiei, Ali Reza Soltanian, and Mojgan Mamani

Title: Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran

Journal: BMC Medical Informatics and Decision Making, Volume 22, Article 192

Publisher: Springer Nature

Year: 2022

DOI: <https://doi.org/10.1186/s12911-022-01939-x>

Article Link: <https://doi.org/10.1186/s12911-022-01939-x>

Task 1: Article Summary

(a) Domain/Application Area and ML Problem

The article belongs to the healthcare and medical decision-support domain. It focuses on predicting mortality among COVID-19 patients using demographic, clinical, and laboratory information.

The specific machine-learning problem is a binary classification problem: Class 0 represents recovered patients and Class 1 represents deceased patients. The researchers wanted not only to predict mortality but also to identify the most influential factors associated with death. The study therefore emphasizes interpretable machine learning, which is especially important in healthcare because physicians need to understand why a model produces a particular prediction.

(b) Key Objective and Research Gap

The main objective of the study was to develop an accurate and interpretable machine-learning model for classifying the mortality risk of COVID-19 patients. The researchers recognized that many previous studies had used machine-learning algorithms to predict COVID-19 mortality, but several of them mainly reported prediction performance without sufficiently explaining which clinical and laboratory factors were responsible for the predictions. This created a research gap because medical professionals require understandable decision rules rather than only a numerical prediction. To address this gap, the authors first used Random Forest to determine the relative importance of 25 demographic, clinical, and laboratory features. Features with more than 6% relative importance were then selected. Four decision-tree approaches—Logistic Model Tree (LMT), C4.5, C5.0, and CART—were developed and compared using recall, accuracy, and F1-score. CART was ultimately selected as the best-performing model and converted into an interpretable decision tree. The resulting tree identified age, gender, BUN, PTT, SGOT, and ESR as important variables and provided understandable rules that could assist clinicians in prioritizing high-risk patients.

Task 2: Methodology Analysis

(a) ML Techniques Used

The study uses a combination of Random Forest and Decision Tree algorithms.

1. Random Forest Feature Selection: The researchers initially considered 25 demographic, clinical, and laboratory features. Random Forest was used to calculate the relative importance of these features. The Random Forest model consisted of 1,000 trees and used the Gini criterion for calculating feature importance. Features having relative importance greater than 6% were selected for the next stage.

The six selected features were: Gender, Age, Blood Urea Nitrogen (BUN), Partial Thromboplastin Time (PTT), Serum Glutamic-Oxaloacetic Transaminase (SGOT), and Erythrocyte Sedimentation Rate (ESR).

2. Logistic Model Tree (LMT): LMT combines decision-tree structures with logistic regression. Logistic regression models are fitted at tree nodes, while LogitBoost is used during model construction.

3. C4.5: C4.5 is an extension of ID3. It uses Information Gain Ratio as its splitting criterion and includes pruning to reduce unnecessary branches and overfitting.

4. C5.0: C5.0 is an improved version of C4.5. It can create smaller trees and is designed to improve computational efficiency and classification performance.

5. CART: CART stands for Classification and Regression Trees. In this study it was used for binary classification. CART creates binary splits using criteria such as the Gini index and can be pruned using cost-complexity pruning.

Dataset Description

The dataset contained information from 2,470 COVID-19 patients admitted to Sina and Besat hospitals in Hamadan, western Iran. The data were collected between February 2020 and July 2021 from medical records. Patients had positive RT-PCR results.

The original dataset contained 25 demographic, clinical, and laboratory features. Examples include age, gender, marital status, smoking, diabetes, hypertension, cardiovascular disease, lung disease, cancer, neurological disease, ESR, BUN, blood sugar, CPK, SGOT, PTT, platelets, sodium, LDH, and PMN.

The target variable was Dead = 1 and Recovered = 0. There were 1,853 recovered patients (75.02%) and 617 deceased patients (24.98%).

For data preparation, records containing missing feature values were removed after examining the missing-data mechanism. Records containing outliers or illogical values were also removed. Random Forest feature selection was then

applied, and only features with relative importance above 6% were retained. The data were randomly divided into training and testing subsets. This splitting process was repeated 10 times to reduce the influence of a single random train-test split, and mean performance measures were reported.

(b) Mapping to CO 4 Topics

CO 4 Topic	Connection with the Article
Inductive Learning	The models learn classification rules from previously observed COVID-19 patient data and use these rules to classify unseen patients.
Decision Trees	LMT, C4.5, C5.0, and CART are directly used for mortality classification.
Statistical Learning	LMT incorporates logistic regression, while Random Forest is used for statistical feature-importance estimation.
Reinforcement Learning	Not directly used in this research.

Methodological Strength

A major strength is interpretability. The final CART model produces understandable decision rules instead of acting as a completely black-box model. This is particularly useful in healthcare because doctors can examine the variables and conditions leading to a high-risk prediction. The researchers also compared multiple tree algorithms instead of relying on only one model.

Methodological Limitation

A major limitation is limited geographic and institutional diversity. The data came from only two hospitals in Hamadan, Iran. Therefore, the model may not generalize equally well to patients from other hospitals, countries, ethnic groups, healthcare systems, or later COVID-19 variants. Another limitation is that some potentially useful laboratory features were excluded because of missing information.

Task 3: Results and Critical Evaluation

(a) Key Results

Model	Recall	F1-Score	Accuracy
LMT	0.9261	0.8573	0.7689
C4.5	0.9189	0.8608	0.7770
C5.0	0.9261	0.8574	0.7689
CART	0.9550	0.8681	0.7824

CART achieved the highest test recall, F1-score, and accuracy among the four compared models. Its test-set performance was Recall = 0.9550, F1-score = 0.8681, and Accuracy = 0.7824. For the total dataset, CART achieved Recall = 0.9385, F1-score = 0.8667, and Accuracy = 0.7834. The reported AUC for CART was 79.5%.

The final CART model contained 35 nodes, consisting of one root node, 16 internal nodes, and 18 leaf nodes. The tree was used to provide interpretable mortality-risk rules. One reported branch indicated that a patient above 60 years of age with BUN greater than 20 and SGOT greater than 45 had an estimated mortality risk of approximately 62.3%.

(b) Critical Evaluation

The results are plausible, but the interpretation requires caution. The 95.5% recall is attractive for a medical mortality-prediction application because missing a high-risk patient can be more serious than incorrectly identifying a low-risk patient as high-risk. However, overall accuracy is only approximately 78.2%, and AUC is 79.5%. Therefore, the model should not be interpreted as an extremely accurate universal mortality predictor.

Potential sources of bias include geographic bias because data were obtained from hospitals in one Iranian region; institutional bias because only two hospitals were used; class imbalance because 75.02% recovered while 24.98% died; missing-feature bias because some laboratory variables were excluded; and temporal bias because COVID-19 variants and treatment protocols changed over time.

Repeating the random train-test split 10 times is useful, but a stronger evaluation would involve external validation using an independent hospital or geographical region. Additional useful metrics would include precision, specificity, balanced

accuracy, Matthews correlation coefficient, calibration score, and confidence intervals.

Additional Experiments

Perform external validation using another hospital.

Use stratified k-fold cross-validation.

Compare CART against Random Forest, XGBoost, SVM, Naive Bayes, and Logistic Regression.

Conduct subgroup analysis by age and gender.

Evaluate model calibration.

Test the model on data from a different time period.

Use explainability techniques such as SHAP to compare feature importance.

Investigate whether incorporating additional laboratory features improves performance.

(c) Real-World Applicability

The proposed technique can be deployed as a clinical decision-support system. A hospital information system could automatically receive patient information and laboratory values, pass them to the CART model, generate a mortality-risk category, and produce a clinical alert.

Potential applications include emergency-room triage, ICU admission prioritization, COVID-19 mortality-risk assessment, hospital resource allocation, early identification of high-risk patients, and supporting physician decision-making.

Challenges in industrial scaling include data availability, inconsistent electronic medical record systems, missing or incorrect medical data, generalization across hospitals, privacy protection, ethical concerns, model drift, and regulatory approval.

Task 4: Reflection and Learning Outcome

(a) Three Key Insights

Insight 1: Decision Trees Provide Interpretable Machine Learning. A decision tree does not simply produce a prediction; it can show the sequence of decisions used

to reach that prediction. Connection to CO 4: This directly relates to Decision Trees and Inductive Learning. The model learns rules from training examples and represents them as a tree structure.

Insight 2: Feature Selection Can Improve Interpretability. The researchers initially considered 25 variables but used Random Forest to select six important features before constructing the final decision trees. Connection to CO 4: This demonstrates the relationship between statistical learning and inductive learning. Feature selection reduces unnecessary information and creates a simpler model that is easier to understand.

Insight 3: Accuracy Alone Is Not Enough. The study reports accuracy, recall, F1-score, and AUC rather than relying on accuracy alone. This is particularly important because the dataset contains substantially more recovered patients than deceased patients. Connection to CO 4: This reinforces the importance of statistical model evaluation and classification metrics.

(b) Proposed Novel Extension

I would extend the research by developing an explainable ensemble model combining CART, Random Forest, and XGBoost, followed by validation on COVID-19 datasets from multiple hospitals and geographical regions.

The first stage would use the same six important variables identified by the original research. Additional clinical and laboratory features would then be evaluated using feature-selection techniques. The models would be trained using stratified cross-validation and tested on an independent hospital dataset. SHAP-based explanations could be added to determine how individual features influence each patient's prediction.

This extension is feasible because the original study already demonstrates that tree-based methods can classify mortality using routine demographic and laboratory information. Ensemble tree algorithms are also computationally practical for structured medical datasets.

The expected impact includes higher predictive reliability, better generalization across hospitals, more robust feature importance, better understanding of

individual predictions, reduced risk of relying on one decision tree, and stronger evidence for real-world clinical deployment.

Overall Critical Review

The article provides a strong example of applying Decision Trees and Inductive Learning to a real-world healthcare classification problem. Its major contribution is not simply obtaining a prediction but producing an interpretable decision structure that can help clinicians understand important mortality-related factors. The combination of Random Forest feature selection with LMT, C4.5, C5.0, and CART provides a useful comparative methodology.

CART produced the best reported test performance, with an accuracy of 78.24%, F1-score of 86.81%, recall of 95.50%, and AUC of 79.5%. However, the results should not be generalized directly to all COVID-19 populations because the dataset was collected from only two hospitals in Hamadan. External validation, larger multi-center datasets, calibration analysis, and comparison with modern ensemble methods would make the conclusions stronger.

Overall, the research demonstrates how inductive learning and decision-tree techniques can transform clinical data into interpretable classification rules, making it a suitable and relevant article for CO 4

Reference

Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran. *BMC Medical Informatics and Decision Making*, 22, 192. <https://doi.org/10.1186/s12911-022-01939-x>